RAJALAKSHMI ENGINEERING COLLEGE

(An Autonomous Institution)

Rajalakshmi Nagar, Thandalam – 602 105

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING (CYBER SECURITY)

CR23532 – CYBER AND DIGITAL FORENSICS

LABORATORY RECORD

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EXPERIMENT 1 : STUDY OF COMPUTER FORENSICS AND DIFFERENT TOOLS USED FOR FORENSIC INVESTIGATION

AIM:

To study the concept of computer forensics, its importance, and the different tools used for forensic investigation.

PROCEDURE:

1. Study the basic concept of computer forensics and its role in investigating cybercrimes.
2. Understand the major steps involved in a forensic investigation, such as identification, collection, preservation, examination, analysis, and reporting of digital evidence.
3. Identify different sources of digital evidence, including hard disks, USB drives, computers, emails, and system logs.
4. Study the different tools used in computer forensic investigations.
5. Understand the functions and applications of each forensic tool.
6. Compare the tools based on their features and purposes.
7. Record the observations and summarize the importance of computer forensics in digital investigations.

Computer forensics is a branch of digital forensics that involves collecting, preserving, examining, and analyzing digital evidence from computers and storage devices. It helps investigators identify cybercrimes and reconstruct digital activities.

The following tools are commonly used in forensic investigations:

S.No	Forensic Tool	Purpose
1	Autopsy	Analyzes hard disks, files, deleted files, and digital evidence.
2	FTK Imager	Creates forensic images and previews digital evidence.
3	Encase Forensic	Supports the acquisition, examination, and analysis of digital evidence.
4	Wireshark	Captures and analyzes network traffic.
5	Volatility	Analyzes memory dumps to investigate running processes and other memory artifacts.
6	Sleuth Kit	Provides command-line tools for examining disk images and file systems.
7	Magnet AXIOM	Helps examine digital evidence from computers, mobile devices, and other sources.

The tools serve different purposes. Some focus on disk analysis, while others are used for memory analysis, network investigation, or digital evidence acquisition. Investigators select tools based on the type of evidence and the requirements of the case.

Result:

Thus, the concept of computer forensics and the different tools used for forensic investigation were studied. The importance of digital evidence collection, preservation, examination, and analysis in investigating cybercrimes was understood.

EXPERIMENT 2 : LIVE FORENSIC INVESTIGATION USING AUTOPSY

AIM:

To perform a basic forensic investigation of a Windows system using Autopsy and identify files, user activity, browser history, deleted files and other digital evidence.

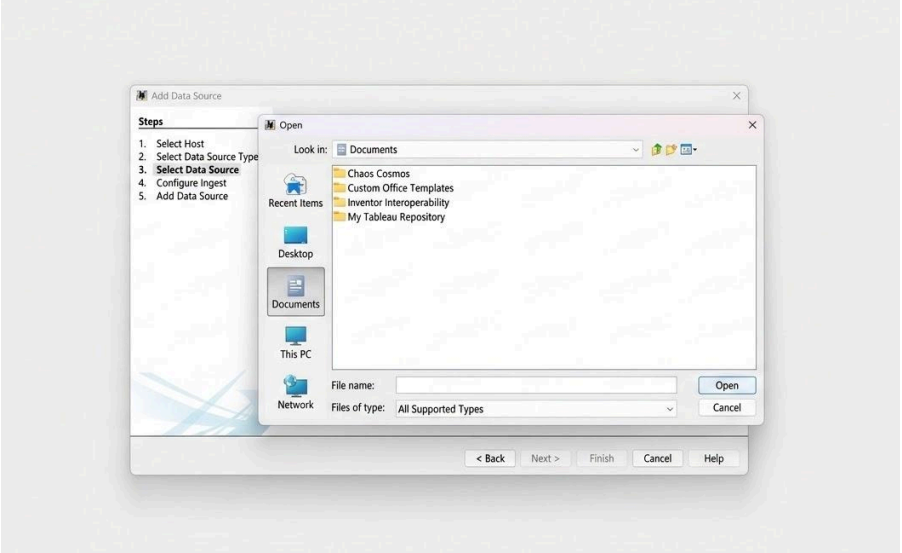
PROCEDURE:

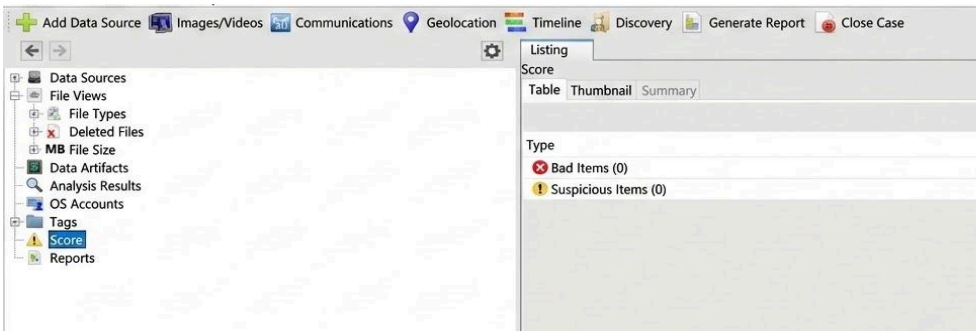
1. Download and Install Autopsy : Download Autopsy from the official website and install it on your PC by following the installation instructions.
2. Create a New Case : Launch Autopsy and select New Case. Enter the case name, base directory, and report output location.
3. Enter Case Details : Enter the case number, examiner name, and other required case details. Click Finish to create the case.
4. Add a Data Source : Select Add Data Source and choose the appropriate data source type, such as a disk image, local disk, or logical files. Browse to the evidence location and select the required source.
5. Configure Ingest Modules : Select the required ingest modules to analyze the evidence. These modules can extract file information, operating system details, user accounts, web history, downloads, cookies, email addresses, and other digital artifacts.
6. Start the Investigation : Click Finish to begin processing the selected data source. Autopsy analyzes the evidence and stores the results in the case database.
7. Examine the Investigation Results : After processing is complete, review the available results in the Autopsy interface, including:
    - Devices Attached: Examine artifacts related to connected devices, if available.
    - EXIF Metadata: Inspect image metadata, such as camera details and timestamps.

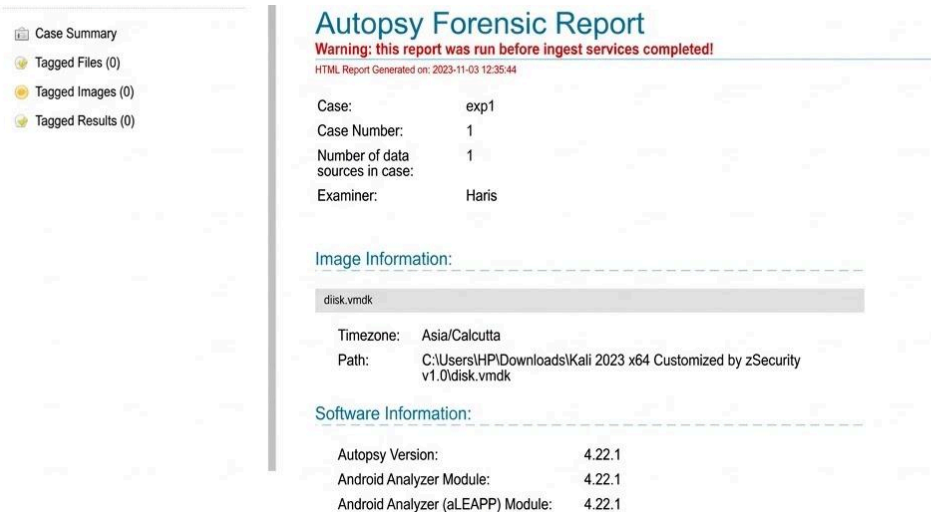
- Installed Programs: Identify installed software, where the relevant artifacts are available.
- Operating System Information: Examine operating system details.
- Operating System User Accounts: Identify user accounts found in the evidence.
- Recent Documents: Find artifacts related to recently accessed documents.
- Web Bookmarks and History: Examine browsing activity and saved bookmarks.
- Web Downloads and Search History: Review download records and search-related artifacts, if available.
- Web Cookies: Analyze cookies recovered from browser data.
- Email Addresses: Identify email addresses found during artifact extraction.

8. Generate the Investigation Report : Review the relevant findings and generate an investigation report using Autopsy's reporting options. Save the report for documentation and further analysis.

OUTPUT:




Add Data Source
Images/Videos
Communications
Geolocation
Timeline
Discovery
Generate Report
Close Case
Data Sources
File Views
File Types
Deleted Files
MB File Size
Data Artifacts
Analysis Results
OS Accounts
Tags
Score
Reports
Listing
Score
Table
Thumbnail
Summary
Type
Bad Items (0)
Suspicious Items (0)


Case Summary
Tagged Files (0)
Tagged Images (0)
Tagged Results (0)
Autopsy Forensic Report
Warning: this report was run before ingest services completed!
HTML Report Generated on: 2023-11-03 12:35:44
Case: exp1
Case Number: 1
Number of data sources in case: 1
Examiner: Haris
Image Information:
diisk.vmdk
Timezone: Asia/Calcutta
Path: C:\Users\HP\Downloads\Kali 2023 x64 Customized by zSecurity v1.0\disk.vmdk
Software Information:
Autopsy Version: 4.22.1
Android Analyzer Module: 4.22.1
Android Analyzer (aLEAPP) Module: 4.22.1


Email Parser Module 4.2.2.1
Embedded File Extractor Module 4.2.2.1
Encryption Detection Module 4.2.2.1
Extension Mismatch Detector Module 4.2.2.1
File Type Identification Module 4.2.2.1
GFX Parser Module 1.2
Hash Lookup Module 4.2.2.1
Interesting Files Identifier Module 4.2.2.1
Keyword Search Module 4.2.2.1
PhotoRec Carver Module 7.0
Picture Analyzer Module 4.2.2.1
Recent Activity Module 4.2.2.1
Virtual Machine Extractor Module 4.2.2.1
YARA Analyzer Module 4.2.2.1
iOS Analyzer (iLEAPP) Module 4.2.2.1
Ingest History:
Job 1
Data Source: disk.vmdk
Status: STARTED
Enabled Modules: Recent Activity
Hash Lookup
File Type Identification
Extension Mismatch Detector
Embedded File Extractor
Picture Analyzer
Keyword Search
Email Parser
Encryption Detection
Interesting Files Identifier
Central Repository
PhotoRec Carver
Virtual Machine Extractor
Data Source Integrity
Android Analyzer (iiLEAPP)
DJI Drone Analyzer
YARA Analyzer
iOS Analyzer (iLEAPP)

RESULT:

The Windows test system was successfully investigated using Autopsy. Relevant digital evidence including user files, browser activity, deleted files and file metadata was identified, examined and documented. A forensic HTML report was successfully generated.

EXPERIMENT 3 : FTK IMAGER - DISK IMAGING AND EVIDENCE ACQUISITION

AIM:

To study and perform disk imaging and digital evidence acquisition using FTK Imager while maintaining the integrity of the original evidence.

PROCEDURE:

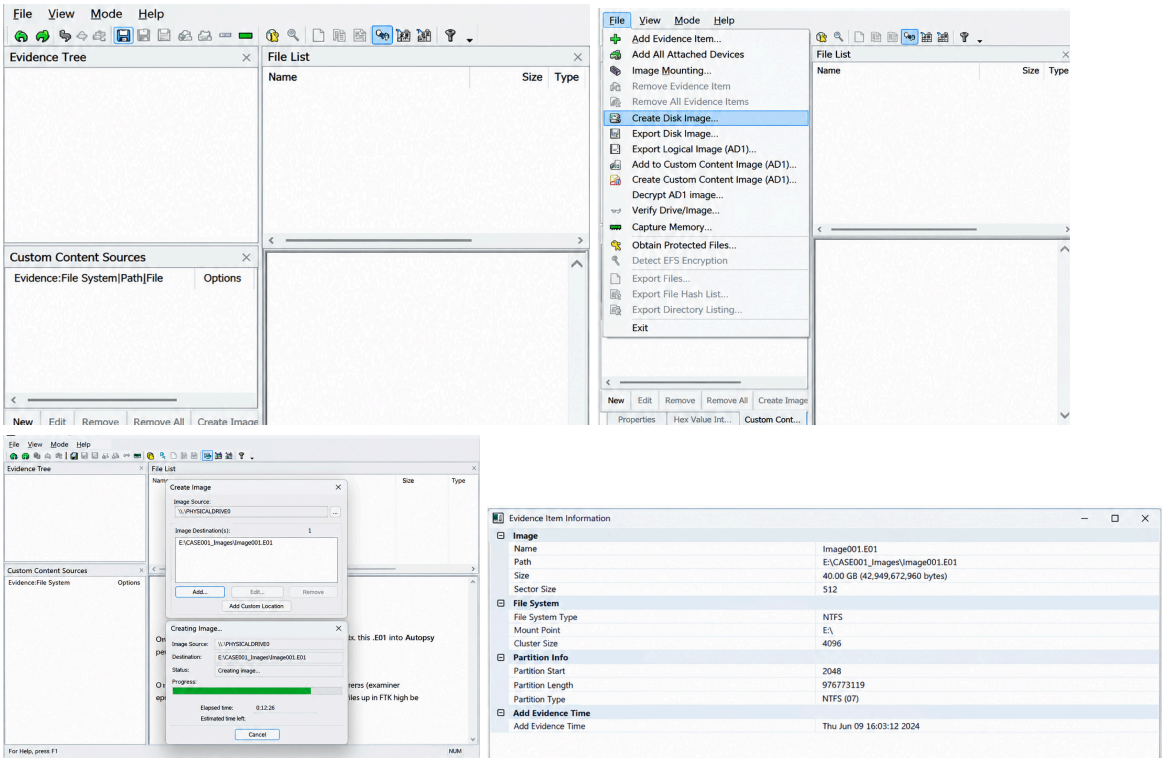
1. Download and install FTK Imager from the official AccessData/Exterro source, if it is not already installed.
2. Open FTK Imager with the necessary permissions.
3. Connect the storage device or select an authorized forensic image file for examination.
4. Select File -> Create Disk Image to begin the disk imaging process.
5. Choose the appropriate source type, such as a physical drive or logical drive, depending on the experiment.
6. Select the source device and verify its details carefully.
7. Choose the required destination image format, such as E01 or Raw (dd).
8. Enter the case number, evidence number, examiner name, and other relevant details when prompted.
9. Specify the destination folder where the forensic image will be saved. Do not save the image to the original evidence device.
10. Start the imaging process and wait until it is completed.
11. If supported by the selected workflow, enable verification and compare the calculated hash values to check the integrity of the acquired image.
12. Record the image details, completion status, and hash values in the investigation notes.

OBSERVATION:

1. FTK Imager is a digital forensic tool used to acquire and preview digital evidence without directly modifying the source during a properly conducted acquisition.

2. The following observations can be recorded during the experiment:
    a. The selected source device and its storage capacity were identified.
    b. The appropriate forensic image format was selected.
3. The disk imaging process created an image file at the specified destination.
4. The image file contained a forensic copy of the selected source data, subject to acquisition limitations.
5. Hash values, such as MD5 or SHA-1, could be calculated to help verify the integrity of the acquired image.
6. Matching hash values for the source and acquired image indicate that their hashed contents are identical for the data covered by the comparison.

OUTPUT:



RESULT:

Thus, disk imaging and digital evidence acquisition using FTK Imager were studied and performed. The process of creating a forensic image and verifying its integrity using hash values was understood.

EXPERIMENT 4 : USB FORENSICS - FINDING THE LAST CONNECTED USB DEVICE ON A SYSTEM

AIM:

To identify the last connected USB storage device on a Windows system by examining USB device history and related system registry entries.

PROCEDURE:

1. Start the Windows computer and log in with the required permissions.
2. Press Windows + R, type regedit, and press Enter to open the Registry Editor.
3. Navigate to the following registry location to examine USB storage device history:
4. HKEY_LOCAL_MACHINE\SYSTEM\CurrentControlSet\Enum\USBSTOR
5. Expand the USBSTOR key to view the USB storage devices previously recognized by the system
6. Select a device entry and examine its subkeys to identify the device model and instance information.
7. Navigate to the following registry location to examine USB device connection information:
8. HKEY_LOCAL_MACHINE\SYSTEM\CurrentControlSet\Enum\USB
9. Record the available device details, such as the manufacturer, product name, device identifier, and serial number, if present.
10. Examine the device properties and relevant timestamps using an appropriate forensic tool, such as USB Detective or a Windows registry analysis tool.
11. If available, examine the Windows SetupAPI log at C:\Windows\INF\setupapi.dev.log for device installation and detection records.
12. Compare the available timestamps and event records to determine

which USB device was most recently connected or installed.

13. Record the findings in the observation table.

Precaution: Registry entries may remain after a USB device is removed. Do not assume that the last registry entry automatically identifies the most recently connected device. Correlate timestamps and other available evidence.

OBSERVATION:

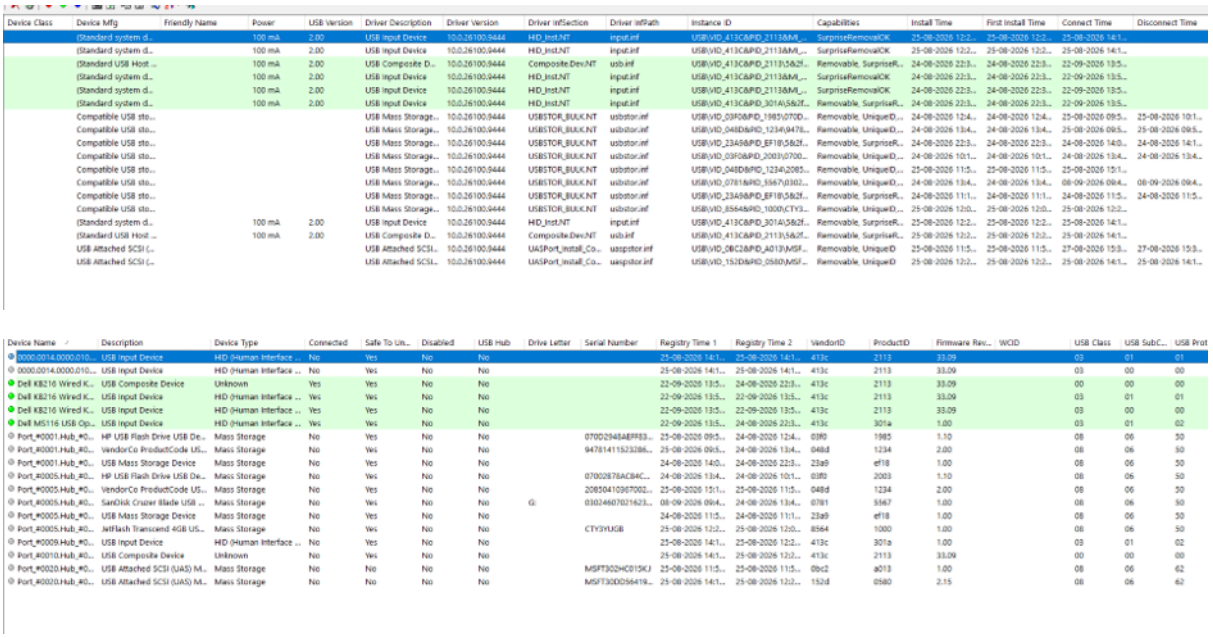
The Windows Registry maintains information about USB devices that have been recognized by the computer. These entries may help investigators identify previously connected USB storage devices.

The following details can be observed during the investigation:

S.No	Parameter	Observation
1	Operating System	Windows
2	Forensic Technique	USB device history analysis
3	Registry Location	USBSTOR
4	Device Information	Manufacturer and model
5	Device Identifier	Serial number or instance ID, if available
6	Connection Evidence	Registry entries and system logs
7	Last Connected Device	Record the device identified from the available evidence
8	Timestamp	Record the relevant date and time, if available

The registry and system logs can provide evidence of previously recognized USB devices. The exact connection time may not be available from every registry entry, so additional logs or forensic artifacts may be required to establish the most recent connection.

OUTPUT:



RESULT:

Thus, USB forensics was studied by examining Windows Registry entries and system logs to identify reviously connected USB storage devices. The available device information and timestamps were analyzed to determine the most recently connected USB device.

EXPERIMENT 5 : VIEWING THE LAST ACTIVITY OF A PC USING WINDOWS FORENSICS

AIM:

To examine the recent activity of a Windows PC using system logs, recent files, and other Windows artifacts for digital forensic investigation.

PROCEDURE:

1. Start the Windows computer and log in with the required permissions.
2. View recent files: Open File Explorer and select Home or Quick access to view recently accessed files, if this feature is enabled.
3. Check recent activity: Press Windows + R, type recent, and press Enter to open the Recent Items folder, if available..
4. Open Event Viewer: Press Windows + R, type eventvwr.msc, and press Enter.
5. Expand Windows Logs and select System to examine system events.
6. Select Windows Logs -> Security to examine available security audit events.
7. Administrative permissions may be required.
8. Select Windows Logs -> Application to view application related events.
9. Review the event date, time, event ID, and description to understand the recorded activity.
10. Check system startup information: In Event Viewer, examine relevant System log
11. events to identify recorded startup and shutdown activity.
12. Check browser activity: Open the browser 's history page to examine recently visited
13. websites, if the history is available and you are authorized to inspect it.
14. Record the relevant activity, timestamps, and observations in the table below.

Precaution: Registry entries may remain after a USB device is removed. Do not

assume that the last registry entry automatically identifies the most recently connected device. Correlate timestamps and other available evidence.

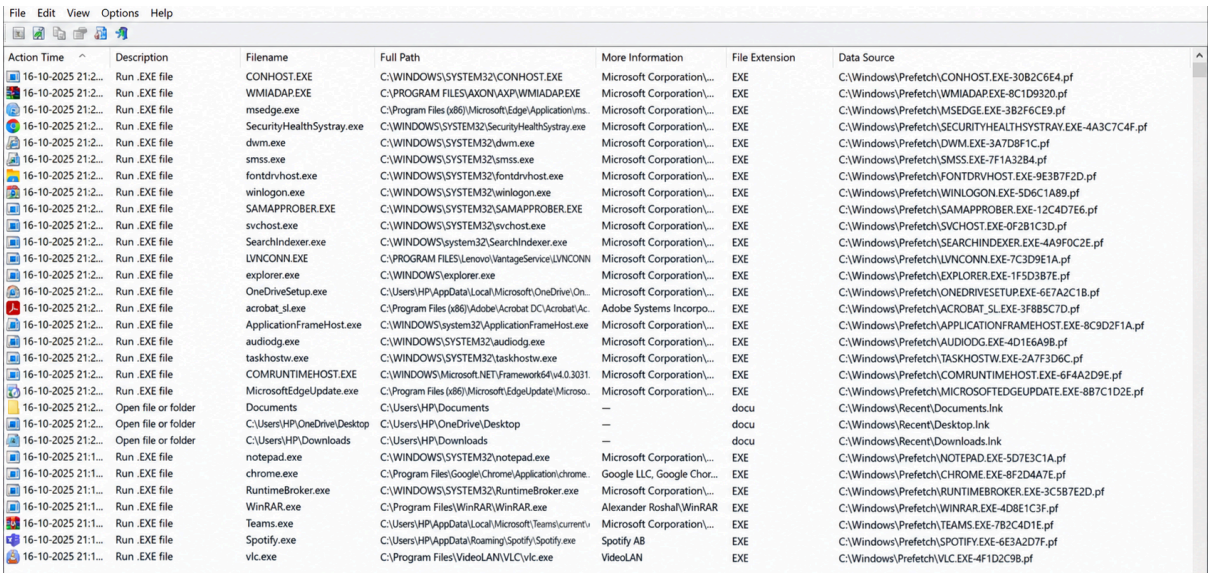
OBSERVATION:

Windows stores different types of activity Information in system logs, recent item records, and application artifacts. These sources can help an investigator understand some of the activities performed on a computer.

S. No.	Activity Source	Observation
1	Recent Items	Recently accessed files or shortcuts, if available
2	Event Viewer – System	System startup, shutdown, and other system events
3	Event Viewer – Security	Recorded login and security events, depending on audit settings
4	Event Viewer – Application	Application-related events and errors
5	Browser History	Recently visited websites, if retained
6	Date and Time	Timestamp associated with the selected event or artifact

The registry and system logs can provide evidence of previously recognized USB devices. The exact connection time may not be available from every registry entry, so additional logs or forensic artifacts may be required to establish the most recent connection.

OUTPUT:



File Edit View Options Help

Action Time	Description	Filename	Full Path	More Information	File Extension	Data Source
16-10-2025 21:2...	Run .EXE file	CONHOST.EXE	C:\WINDOWS\SYSTEM32\CONHOST.EXE	Microsoft Corporation\...	EXE	C:\Windows\Prefetch\CONHOST.EXE-30B2C6E4.pf
16-10-2025 21:2...	Run .EXE file	WMIADAP.EXE	C:\PROGRAM FILES\AXON\AXP\WMIADAP.EXE	Microsoft Corporation\...	EXE	C:\Windows\Prefetch\WMIADAP.EXE-8C1D9320.pf
16-10-2025 21:2...	Run .EXE file	msedge.exe	C:\Program Files (x86)\Microsoft\Edge\Application\ms..	Microsoft Corporation\...	EXE	C:\Windows\Prefetch\MSEDGE.EXE-3B2F6CE9.pf
16-10-2025 21:2...	Run .EXE file	SecurityHealthSystray.exe	C:\WINDOWS\SYSTEM32\SecurityHealthSystray.exe	Microsoft Corporation\...	EXE	C:\Windows\Prefetch\SECURITYHEALTHSYSTRAY.EXE-4A3C7C4F.pf
16-10-2025 21:2...	Run .EXE file	dwm.exe	C:\WINDOWS\SYSTEM32\dwm.exe	Microsoft Corporation\...	EXE	C:\Windows\Prefetch\DWM.EXE-3A7D8F1C.pf
16-10-2025 21:2...	Run .EXE file	smss.exe	C:\WINDOWS\SYSTEM32\smss.exe	Microsoft Corporation\...	EXE	C:\Windows\Prefetch\SMSS.EXE-7F1A32B4.pf
16-10-2025 21:2...	Run .EXE file	fontdrvhost.exe	C:\WINDOWS\SYSTEM32\fontdrvhost.exe	Microsoft Corporation\...	EXE	C:\Windows\Prefetch\FONTDRVHOST.EXE-9E3B7F2D.pf
16-10-2025 21:2...	Run .EXE file	winlogon.exe	C:\WINDOWS\SYSTEM32\winlogon.exe	Microsoft Corporation\...	EXE	C:\Windows\Prefetch\WINLOGON.EXE-5D6C1A89.pf
16-10-2025 21:2...	Run .EXE file	SAMAPPROBER.EXE	C:\WINDOWS\SYSTEM32\SAMAPPROBER.EXE	Microsoft Corporation\...	EXE	C:\Windows\Prefetch\SAMAPPROBER.EXE-12C4D7E6.pf
16-10-2025 21:2...	Run .EXE file	svchost.exe	C:\WINDOWS\SYSTEM32\svchost.exe	Microsoft Corporation\...	EXE	C:\Windows\Prefetch\SVCHOST.EXE-0F2B1C3D.pf
16-10-2025 21:2...	Run .EXE file	SearchIndexer.exe	C:\WINDOWS\system32\SearchIndexer.exe	Microsoft Corporation\...	EXE	C:\Windows\Prefetch\SEARCHINDEXER.EXE-4A9F0C2E.pf
16-10-2025 21:2...	Run .EXE file	LVNCONN.EXE	C:\PROGRAM FILES\Lenovo\VantageService\LVNCONN	Microsoft Corporation\...	EXE	C:\Windows\Prefetch\LVNCONN.EXE-7C3D9E1A.pf
16-10-2025 21:2...	Run .EXE file	explorer.exe	C:\WINDOWS\explorer.exe	Microsoft Corporation\...	EXE	C:\Windows\Prefetch\EXPLORER.EXE-1F5D3B7E.pf
16-10-2025 21:2...	Run .EXE file	OneDriveSetup.exe	C:\Users\HP\AppData\Local\Microsoft\OneDrive\On..	Microsoft Corporation\...	EXE	C:\Windows\Prefetch\ONEDRIVESETUP.EXE-6E7A2C1B.pf
16-10-2025 21:2...	Run .EXE file	acrobat_sl.exe	C:\Program Files (x86)\Adobe\Acrobat DC\Acrobat\Ac..	Adobe Systems Incorpo...	EXE	C:\Windows\Prefetch\ACROBAT_SL.EXE-3F8B5C7D.pf
16-10-2025 21:2...	Run .EXE file	ApplicationFrameHost.exe	C:\WINDOWS\system32\ApplicationFrameHost.exe	Microsoft Corporation\...	EXE	C:\Windows\Prefetch\APPLICATIONFRAMEHOST.EXE-8C9D2F1A.pf
16-10-2025 21:2...	Run .EXE file	audiodg.exe	C:\WINDOWS\SYSTEM32\audiodg.exe	Microsoft Corporation\...	EXE	C:\Windows\Prefetch\AUDIODG.EXE-4D1E6A9B.pf
16-10-2025 21:2...	Run .EXE file	taskhostw.exe	C:\WINDOWS\SYSTEM32\taskhostw.exe	Microsoft Corporation\...	EXE	C:\Windows\Prefetch\TASKHOSTW.EXE-2A7F3D6C.pf
16-10-2025 21:2...	Run .EXE file	COMRUNTIMEHOST.EXE	C:\WINDOWS\Microsoft.NET\Framework64\v4.0.3031..	Microsoft Corporation\...	EXE	C:\Windows\Prefetch\COMRUNTIMEHOST.EXE-6F4A2D9E.pf
16-10-2025 21:2...	Run .EXE file	MicrosoftEdgeUpdate.exe	C:\Program Files (x86)\Microsoft\EdgeUpdate\Microso..	Microsoft Corporation\...	EXE	C:\Windows\Prefetch\MICROSOFTEDGEUPDATE.EXE-8B7C1D2E.pf
16-10-2025 21:2...	Open file or folder	Documents	C:\Users\HP\Documents	—	docu	C:\Windows\Recent\Documents.lnk
16-10-2025 21:2...	Open file or folder	C:\Users\HP\OneDrive\Desktop	C:\Users\HP\OneDrive\Desktop	—	docu	C:\Windows\Recent\Desktop.lnk
16-10-2025 21:2...	Open file or folder	C:\Users\HP\Downloads	C:\Users\HP\Downloads	—	docu	C:\Windows\Recent\Downloads.lnk
16-10-2025 21:1...	Run .EXE file	notepad.exe	C:\WINDOWS\SYSTEM32\notepad.exe	Microsoft Corporation\...	EXE	C:\Windows\Prefetch\NOTEPAD.EXE-5D7E3C1A.pf
16-10-2025 21:1...	Run .EXE file	chrome.exe	C:\Program Files\Google\Chrome\Application\chrome..	Google LLC, Google Chor...	EXE	C:\Windows\Prefetch\CHROME.EXE-8F2D4A7E.pf
16-10-2025 21:1...	Run .EXE file	RuntimeBroker.exe	C:\WINDOWS\SYSTEM32\RuntimeBroker.exe	Microsoft Corporation\...	EXE	C:\Windows\Prefetch\RUNTIMEBROKER.EXE-3C5B7E2D.pf
16-10-2025 21:1...	Run .EXE file	WinRAR.exe	C:\Program Files\WinRAR\WinRAR.exe	Alexander Roshal\WinRAR	EXE	C:\Windows\Prefetch\WINRAR.EXE-4D8E1C3F.pf
16-10-2025 21:1...	Run .EXE file	Teams.exe	C:\Users\HP\AppData\Local\Microsoft\Teams\current\..	Microsoft Corporation\...	EXE	C:\Windows\Prefetch\TEAMS.EXE-7B2C4D1E.pf
16-10-2025 21:1...	Run .EXE file	Spotify.exe	C:\Users\HP\AppData\Roaming\Spotify\Spotify.exe	Spotify AB	EXE	C:\Windows\Prefetch\SPOTIFY.EXE-6E3A2D7F.pf
16-10-2025 21:1...	Run .EXE file	vlc.exe	C:\Program Files\VideoLAN\VLC\vlc.exe	VideoLAN	EXE	C:\Windows\Prefetch\VLC.EXE-4F1D2C9B.pf

The available records depend on Windows settings, user activity, audit policies, and log retention. Recent file entries and browser history may be incomplete or deleted. A recorded event should not automatically be treated as proof of a person's actions.

RESULT:

Thus, the recent activity of a Windows PC was examined using Windows Event Viewer, recent file records, and browser history. The available artifacts and timestamps were studied to understand how computer activity can be investigated using digital forensic techniques.

EXPERIMENT 6 : EXTRACTION OF BROWSER ARTIFACTS FOR DIGITAL FORENSIC INVESTIGATION

AIM:

To extract and examine browser artifacts such as browsing history, cookies, cached files, downloads, and saved browsing information for digital forensic investigation.

PROCEDURE:

1. Start the computer and log in to the Windows system with the necessary permissions.
2. Identify the web browser installed on the system, such as Google Chrome or Microsoft Edge.
3. Open the browser and examine its history by pressing Ctrl + H.
4. Record the available information, including website URLs, page titles, and visit times.
5. Open the browser's download history by pressing Ctrl + J and note the available download details.
6. Locate the browser's user profile directory. For example, Chrome commonly stores profile data under %LOCALAPPDATA%\Google\Chrome\User Data\Default.
7. Identify relevant browser artifact files, such as:

   History: browsing history and download records.

   Cookies: cookie information.

   Web Data : certain autofill and browsing-related information.

   Login Data : saved login information stored by the browser.

   Cache : cached web resources, depending on the browser version.

8. Use a forensic tool, such as Autopsy, to examine a forensic image or

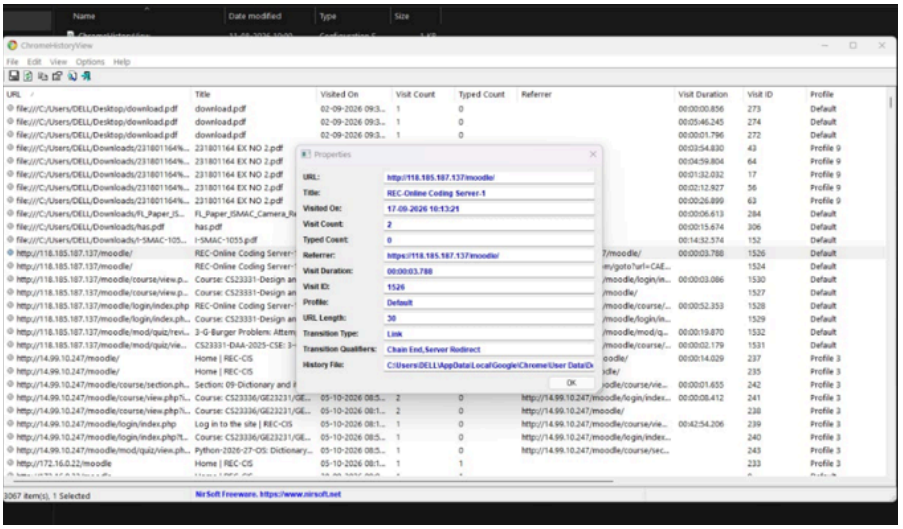
an authorized copy of the browser profile.

9. In Autopsy, create or open a case, add the relevant disk image or data source, and allow the tool to process the available browser artifacts.
10. Review the extracted results, including URLs, timestamps, download information, and other available browser records.
11. Record the findings in the observation table.

Precaution: Use only data you are authorized to examine. Avoid opening or sharing private passwords or sensitive personal information. For forensic work, preserve the original evidence and analyze a verified copy whenever possible.

OBSERVATION:

Windows stores different types of activity Information in system logs, recent item records, and application artifacts. These sources can help an investigator understand some of the activities performed on a computer.



The registry and system logs can provide evidence of previously recognized USB devices. The exact connection time may not be available from every registry entry, so additional logs or forensic artifacts may be required to establish the most recent connection.

OUTPUT:

Browser artifacts provide useful information about a user's web activity. These artifacts can help investigators reconstruct browsing-related events, subject to the information retained by the browser.

S.No	Artifact	Information Obtained
1	Browsing History	Visited URLs, page titles, and available timestamps
2	Download History	Downloaded file names, locations, and related timestamps
3	Cookies	Website-related session and preference information
4	Cache	Cached web resources and related metadata
5	Web Data	Certain autofill and browsing-related records
6	Login Data	Stored login-related records; sensitive values should not be exposed
7	Forensic Tool	Autopsy or another suitable browser artifact analysis tool

The examination may reveal website visits, download activity, and other browsing-related information. The availability of artifacts depends on the browser, its version, user settings, and whether the records have been deleted or overwritten.

RESULT:

Thus, browser artifact extraction and examination were studied using browser history, download records, cookies, cache files, and other available browser data. The importance of browser artifacts in reconstructing digital activity during forensic investigations was understood.

EXPERIMENT 7 : COMPARISON OF TWO FILES FOR FORENSIC INVESTIGATION USING COMPARE IT SOFTWARE

AIM:

To compare two files using Compare It software and identify similarities, differences, and modifications in their contents for digital forensic investigation.

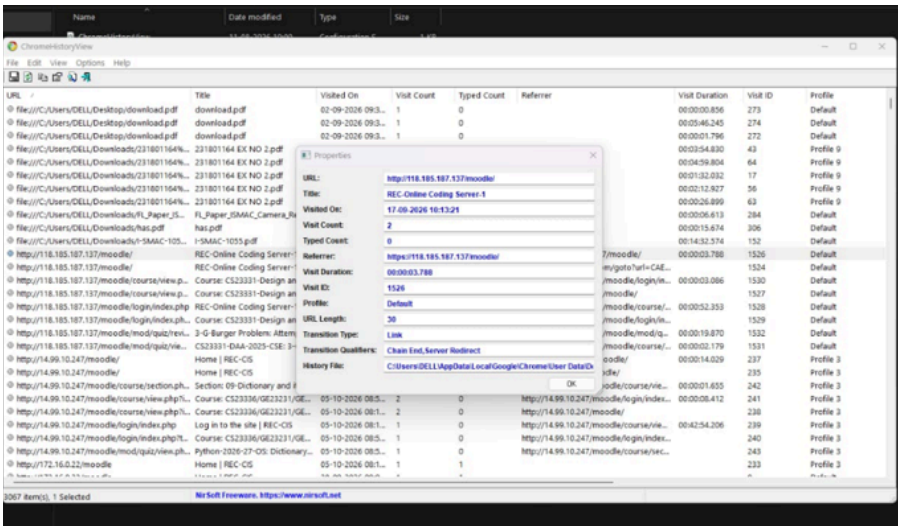
PROCEDURE:

1. Start the computer and open the Compare It software.
2. Prepare two files that need to be compared. The files may contain text, code, or other supported content.
3. Keep both files in an accessible folder and note their names and locations.
4. Open Compare It software.
5. Select the first file as the original or reference file.
6. Select the second file as the file to be compared.
7. Start the comparison using the software's compare option.
8. Observe the differences highlighted by the software, including added deleted, or modified lines of text.
9. Examine the matching sections to identify content that remains unchanged.
10. Record the differences and similarities between the two files.
11. Save or capture the comparison report or screenshots, if the software supports these features.
12. Document the findings for the forensic investigation.

OBSERVATION:

Compare It software helps identify differences between two files by displaying their contents side by side and highlighting changes. This is useful for examining whether

a file has been modified.



The comparison may reveal changes in text, lines, or other supported content. However, a visual comparison alone does not establish when a file was modified or who modified it. Hash comparison can help determine whether two files have identical contents.

OUTPUT:

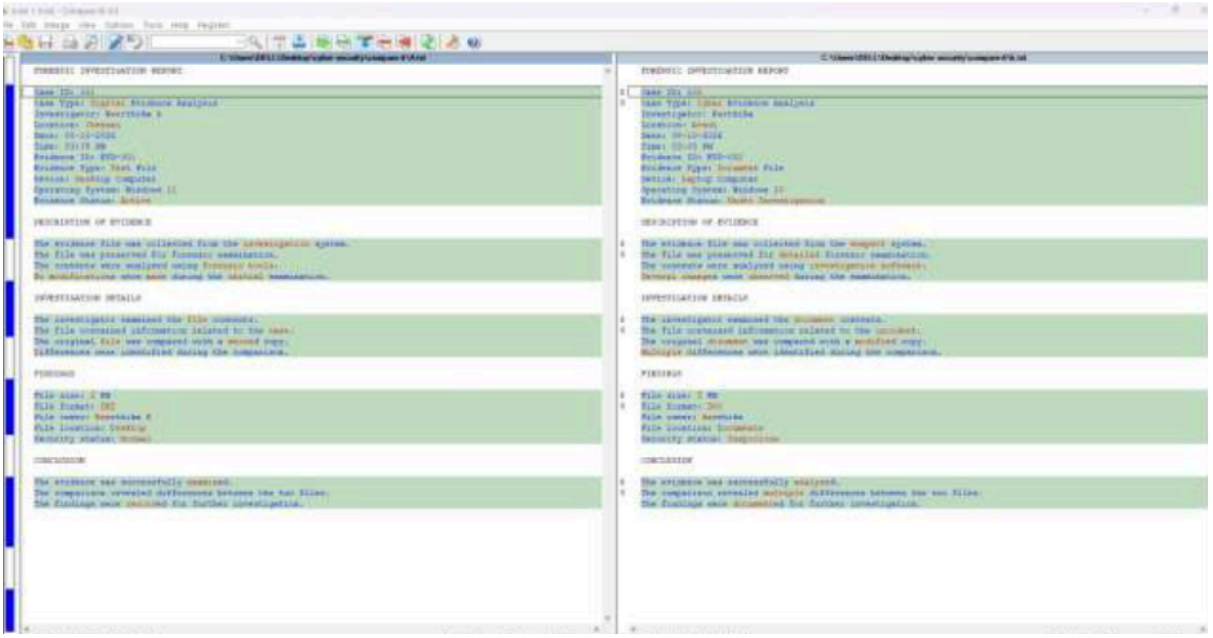
Browser artifacts provide useful information about a user's web activity. These artifacts can help investigators reconstruct browsing-related events, subject to the information retained by the browser.

S.No	Artifact	Information Obtained
1	Browsing History	Visited URLs, page titles, and available timestamps
2	Download History	Downloaded file names, locations, and related timestamps
3	Cookies	Website-related session and preference information
4	Cache	Cached web resources and related metadata
5	Web Data	Certain autofill and browsing-related records
6	Login Data	Stored login-related records; sensitive values should not be exposed

7	Forensic Tool	Autopsy or another suitable browser artifact analysis tool

The examination may reveal website visits, download activity, and other browsing-related information. The availability of artifacts depends on the browser, its version, user settings, and whether the records have been deleted or overwritten.





RESULT:

Thus, two files were compared using Compare It software, and the similarities and differences in their contents were examined. The use of file comparison in identifying possible modifications during digital forensic investigations was understood.

Experiment 8 : Collection of Email Evidence from a Victim's PC

Aim:

To study the methods used to identify, collect, preserve, and examine email evidence from a computer for digital forensic investigation.

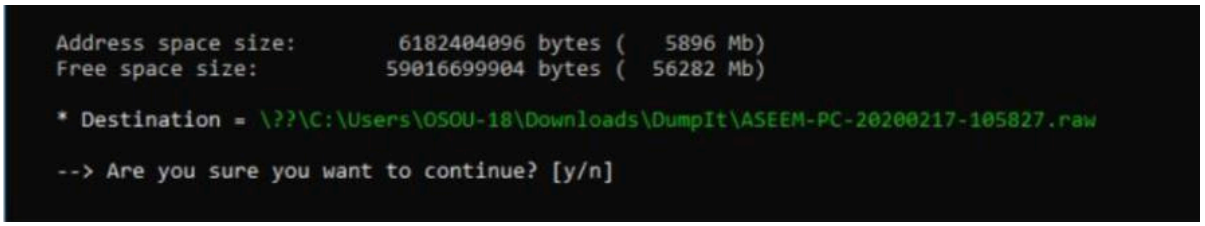
Procedure:

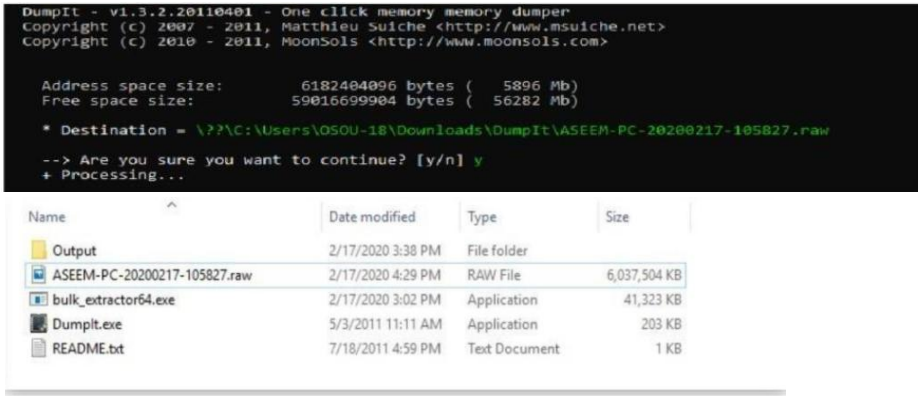
1. Start the computer and obtain proper authorization to examine the email evidence.
2. Identify the email service or client used on the system, such as Gmail, Microsoft Outlook, or Mozilla Thunderbird.
3. Record the system details, date, time, user account information, and other relevant case details.
4. If the email account is accessed through a web browser, examine the available email messages and relevant browser artifacts, following the approved evidence-handling procedure.
5. If a desktop email client is used, identify its local mail storage files. For example, Outlook may use .pst or .ost files, while Thunderbird may store messages in MBOX files or other profile data.
6. Create a forensic image of the storage device or an authorized copy of the relevant email data, using a suitable forensic tool.
7. Preserve the original evidence and perform the examination on a verified copy whenever possible.
8. Export relevant email messages in a suitable format, such as .eml or .msg, where supported.
9. Examine the email sender, recipient, subject, date, time, message body, and available message headers.

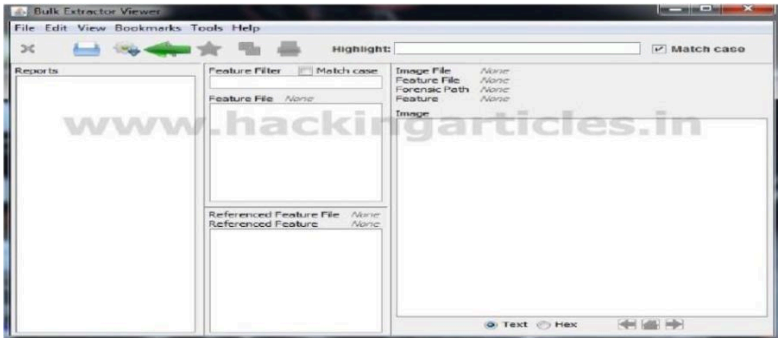
10. Review attachments carefully and record their file names, types, and hash values where appropriate.
11. Calculate cryptographic hash values, such as SHA-256, for collected evidence files and maintain an evidence log.
12. Document the collection method, tools used, timestamps, and findings in the forensic report.

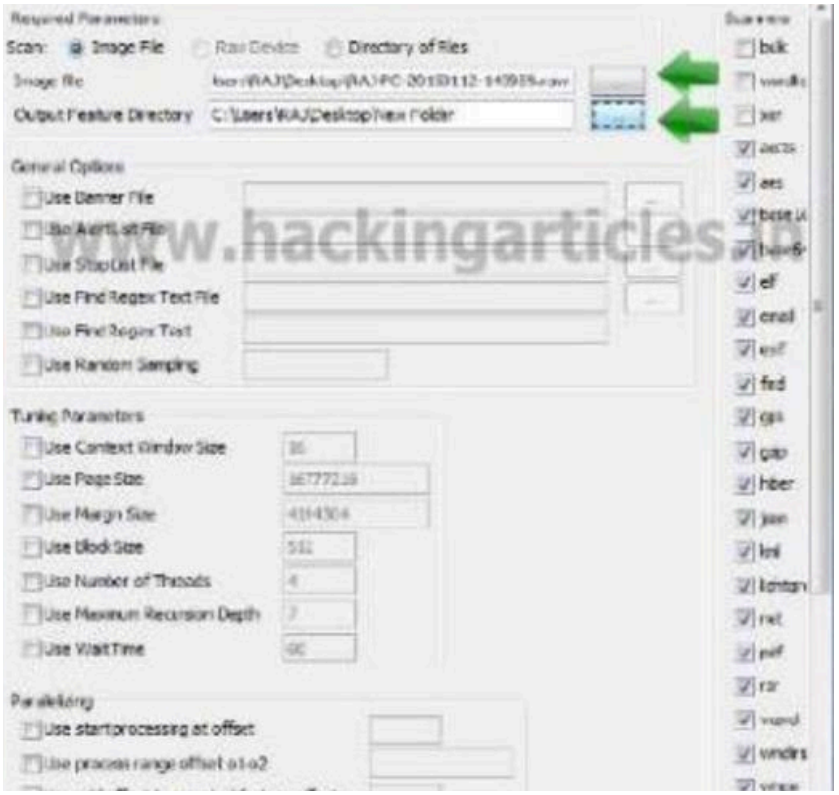
Output:

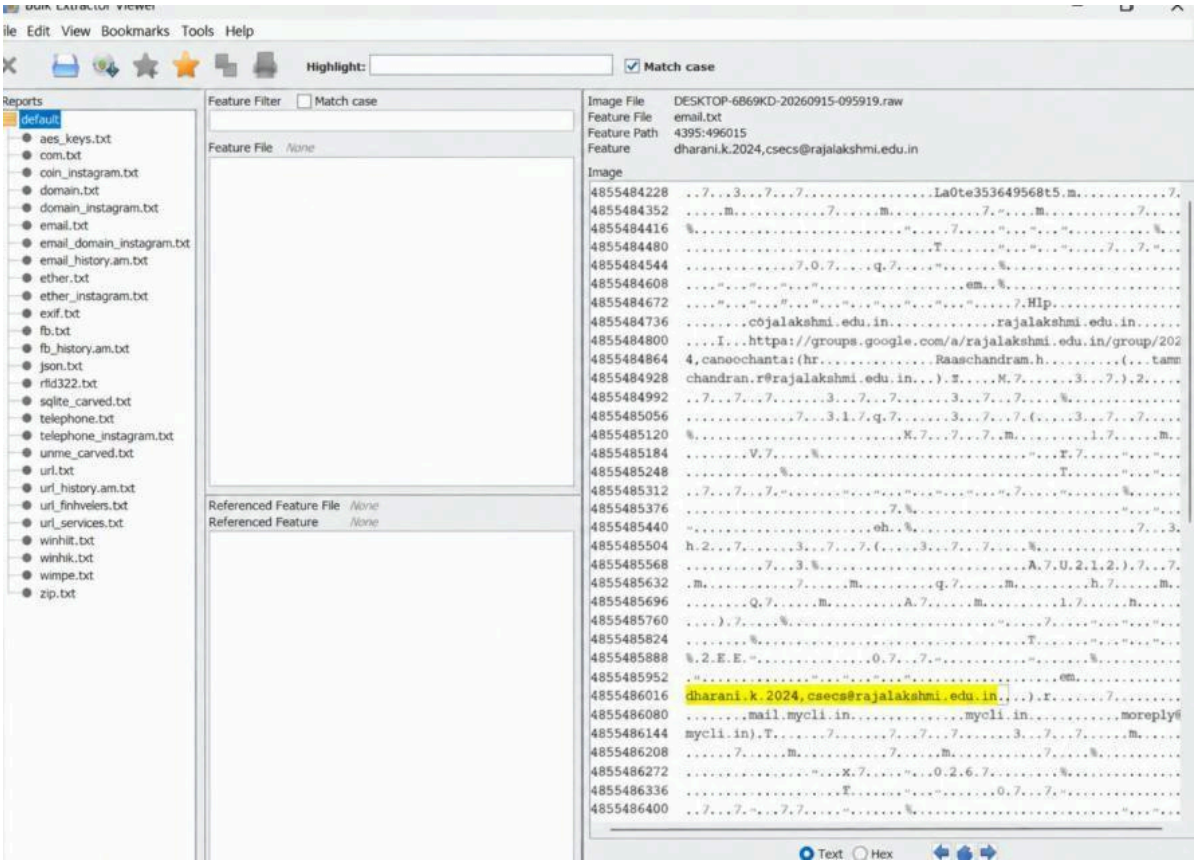







Bulk Extractor Viewer
File Edit View Bookmarks Tools Help
Highlight:
Match case
Reports
Feature Filter
Match case
Feature File
Image File
Feature File
Forensic Path
Feature
Image
Referenced Feature File
Referenced Feature
Text
Hex
www.hackingarticles.in


Scan:
Image File
Raw Device
Directory of Files
Output Feature Directory
C:\Users\RAJ\Desktop\New Folder
General Options
Use Banner File
Use Stop List File
Use Find Regex Text File
Use Random Sampling
Tuning Parameters
Use Context Window Size
Use Page Size
Use Margin Size
Use Block Size
Use Number of Threads
Use Maximum Recursion Depth
Use Wait Time
Use startprocessing at offset
bulk
aes
elf
email
exif
fed
gps
gzip
hiber
jpeg
kml
net
pdf
rar
vmdks
www.hackingarticles.in


File Edit View Bookmarks Tools Help
Highlight:
Match case
Reports
default
aes_keys.txt
com.txt
coin_instagram.txt
domain.txt
domain_instagram.txt
email.txt
email_domain_instagram.txt
email_history.am.txt
ether.txt
ether_instagram.txt
exif.txt
fb.txt
fb_history.am.txt
json.txt
rfid322.txt
sqlite_carved.txt
telephone.txt
telephone_instagram.txt
unme_carved.txt
url.txt
url_history.am.txt
url_finhvelers.txt
url_services.txt
winhlit.txt
winhik.txt
wimpe.txt
zip.txt
Feature Filter
Match case
Feature File None
Referenced Feature File None
Referenced Feature None
Image File DESKTOP-6B69KD-20260915-095919.raw
Feature File email.txt
Feature Path 4395:496015
Feature dharani.k.2024,csecs@rajalakshmi.edu.in
Image
4855484228
4855484352
4855484416
4855484480
4855484544
4855484608
4855484672
4855484736
4855484800
4855484864
4855484928
4855484992
4855485056
4855485120
4855485184
4855485248
4855485312
4855485376
4855485440
4855485504
4855485568
4855485632
4855485696
4855485760
4855485824
4855485888
4855485952
4855486016
4855486080
4855486144
4855486208
4855486272
4855486336
4855486400
dharani.k.2024,csecs@rajalakshmi.edu.in
Text
Hex

Observation:

Email evidence may contain important information about phishing, fraud, unauthorized access, harassment, and other cyber incidents. The evidence must be collected carefully to preserve its integrity.

S. No.	Evidence Type	Information Obtained
1	Sender and Recipient	Email addresses involved in communication
2	Subject	Purpose or topic of the email
3	Date and Time	Available message timestamps
4	Email Header	Routing details and technical metadata
5	Message Body	Content of the communication
6	Attachments	Files shared through the email
7	Local Mail Files	Stored email data from the email client
8	Hash Value	Used to verify whether collected files have changed
9	Evidence Log	Details of collection and handling

Email evidence may be available in a mail server, webmail account, local email client, browser artifacts, or backups. The exact information depends on the email service and the data retained by the system.

Result:

Thus, the methods for collecting and preserving email evidence from a computer were studied. The importance of examining email headers, message content, attachments, and metadata while maintaining evidence integrity was understood.

EXPERIMENT 9 : HIDING AND EXTRACTING TEXT FILES BEHIND AN IMAGE USING COMMAND PROMPT

AIM:

To hide a text file behind an image file and extract the hidden text file using Windows Command Prompt as a basic digital forensic experiment.

PROCEDURE:

Part A: Hiding a Text File Behind an Image

1. Create a folder named Forensics on the Desktop.
2. Copy an image file named image.jpg into the folder.
3. Create a text file named secret.txt in the same folder and enter some sample text.
4. Open Command Prompt by pressing Windows + R, typing cmd, and pressing Enter.
5. Navigate to the folder by entering:

```
cd %USERPROFILE%\Desktop\Forensics
```

6. Combine the image and text files using the following command:

```
copy /b image.jpg + secret.txt hidden.jpg
```

7. A new file named hidden.jpg will be created containing the bytes of both files.

Part B: Extracting the Hidden Text File

8. In Command Prompt, execute the following command:

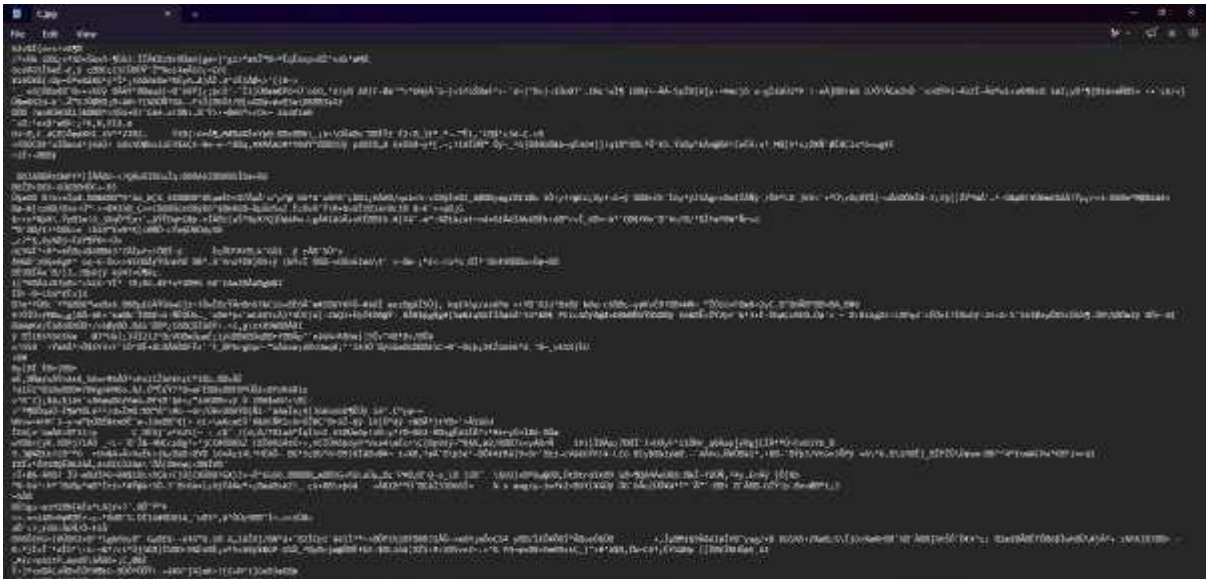
```
find /v /c "" hidden.jpg
```

This command counts lines and does not extract the hidden file; it is not required for extraction.

9. To recover the appended text, use a binary-aware extraction method or a suitable forensic utility that can locate the appended data.
10. For a simple demonstration when the appended text is plain text, open hidden.jpg with a text editor and inspect the end of the file. The text may be visible, but this is not a reliable method of extracting it.
11. Compare the recovered text with the original secret.txt and record the result.

Note: The copy /b command appends the text file to the image; it does not encrypt the text or conceal it securely. The resulting file may still open as an image, depending on the image format and viewer.

OUTPUT:



RESULT:

Thus, a text file was appended to an image file using Windows Command Prompt. The resulting file was examined to understand how extra data can be stored after image data and how such appended content may be detected during digital forensic investigation.

Experiment 10 - Extraction of EXIF Data from an Image File using EXIF Reader Software

AIM:

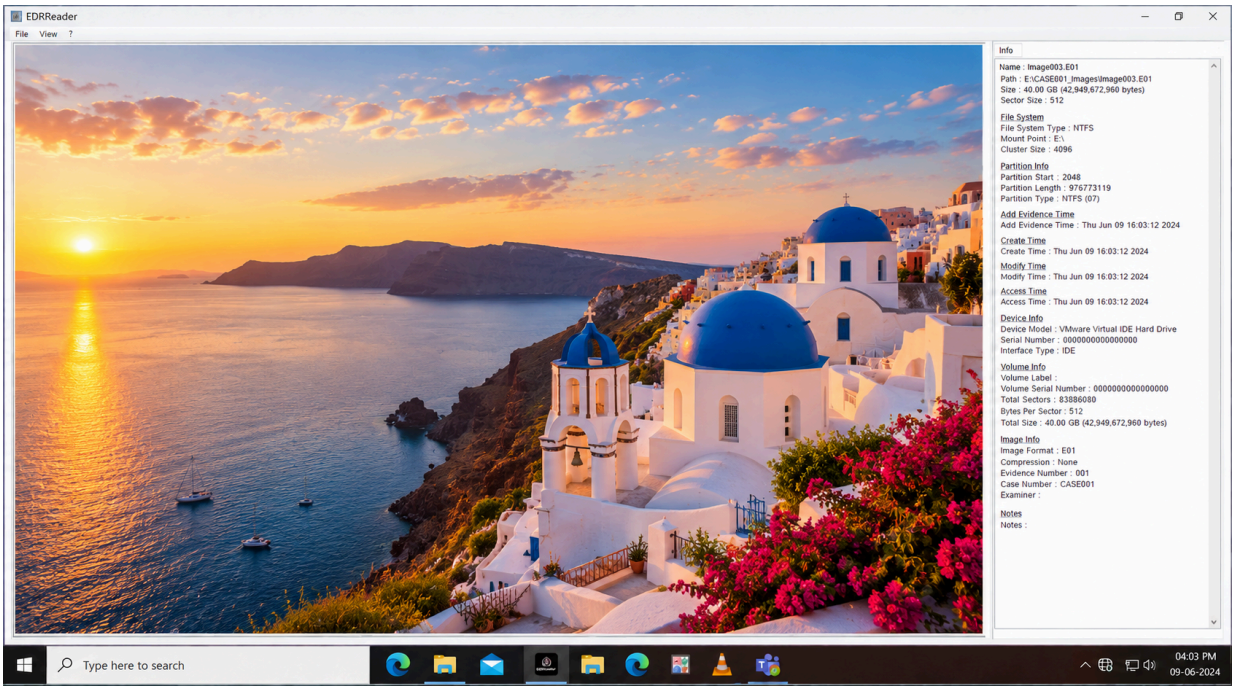
To extract and examine Exchangeable Image File Format (EXIF) metadata from an image file using EXIF Reader software for digital forensic investigation.

PROCEDURE:

1. Start the computer and locate an image file, such as a JPEG photograph, for examination.
2. Install or open a suitable EXIF Reader software application.
3. Launch the software and select the option to open or browse for an image file.
4. Navigate to the folder containing the image and select the required file.
5. Load the image into the EXIF Reader software.
6. Allow the software to analyze the image and display its available EXIF metadata.
7. Examine the metadata fields, including camera manufacturer, camera model, date and time, image dimensions, and orientation.
8. Check whether GPS coordinates or location information are present in the image metadata.
9. Record the available metadata values in the observation table.
10. If supported, export or save the metadata report for documentation.
11. Close the software after recording the findings.

OBSERVATION:

EXIF metadata is information embedded in an image file by a camera, smartphone, or image-processing application. It can provide useful details about how and when an image was created.



The following information may be obtained using EXIF Reader:

. No.	Metadata Field	Information Obtained
1	File Name	Name of the examined image
2	File Format	JPEG, TIFF, or another supported format
3	Camera Manufacturer	Camera or device manufacturer, if available
4	Camera Model	Model of the device, if recorded

5	Date and Time	Recorded image capture date and time, if available
6	Image Dimensions	Width and height in pixels
7	Orientation	Orientation information recorded in the metadata
8	GPS Information	Latitude and longitude, if present
9	Software Information	Software used to process the image, if recorded

The availability of EXIF data depends on the image format, device, and processing history. Some applications remove metadata when an image is edited or shared. Recorded timestamps may also be inaccurate or affected by incorrect device settings.

RESULT:

Thus, EXIF metadata was extracted and examined from an image file using EXIF Reader software. The importance of image metadata, including device details, timestamps, image properties, and possible location information, in digital forensic investigations was understood.